

SCX

SCX

July 2, 2026

Abstract

SCXState-Conditioned eXpertise — Robbins-MonroSpring SE-1Chernoff-Hoeffding The-
 orem $1I(Y; P \mid S) > 0$ —
 SCX “” — $\Delta_s > 0$ bug Spring SE-1 t_{crit} $\text{PLO}(1/t)$ “” SCX Theorem 1 — F_1 1 —
 $\exp(-2M\Delta_s^2)$ $\mathcal{S}_i$ $d_i\kappa_i\text{PL}\mu_i$ $\theta_i^*(t)$ “” — $\Omega(\max_i d_i)$
 “”
 SCX Robbins-MonroSpring

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1.1 SCX

SCX[?, ?]State Crystallization SitusSpringYajieCercis

- (1) $\mathcal{S} = \{s_1, \dots, s_N\}N$ “” $s_i \in \mathbb{R}^{d_s}/$
- (2) $\Delta_s(t) = p_{\text{noisy},s}(t) - p_{\text{clean},s}(t)$ stM $\Delta_s > 0$
- (3) **SpringSE-1** θ_t Robbins-Monro

$$\theta_{t+1} = \theta_t - \alpha_t \cdot \nabla \mathcal{L}_{\text{Spring}}(\theta_t; \mathcal{B}_t) + \xi_t \tag{1}$$

$$\alpha_t \sum \alpha_t = \infty \sum \alpha_t^2 < \infty \xi_t$$

- (4) **Theorem 1** MF_1

$$F_1 \geq 1 - \frac{1}{\eta} \sum_{s \in \mathcal{S}} \rho_s \cdot \exp(-2M\Delta_s^2) \tag{2}$$

- (5) PSY $I(Y; P \mid S) > 0$ SCX

1.2

— Spring $\theta_t\mathcal{S}$ Yajie $\{E_m\}_{m=1}^M\Delta_s(t)$ SCX normative force
 — structural isomorphism ??

1.3

“—”SCX

Table 1: SCX		
SCX		
s_i	$\mathbb{R}^{d_s}$	/
Spring θ_t	$\mathbb{R}^{ \theta }$	
$\Delta_s(t)$	$[-1, 1]$	s
SE-1	$\theta_{t+1} = \theta_t - \alpha_t \nabla \mathcal{L}$	/
$\mathcal{M}_t$	$\{(s_i, \text{score}_t(s_i))\}$	
E_m	$\mathcal{X} \rightarrow \{0, 1\}$	/
Cercis	$[0, 1]$	

2

2.1

Definition 2.1 (——SCX). *Spring* $t' > t$ $\theta_{t'} = \theta_t$ $\mathcal{M}_{t'} = \mathcal{M}_t$ $s \Delta_s \tau \rightarrow \infty$

$$\boxed{\lim_{\tau \rightarrow \infty} \Delta_s(t + \tau) = 0} \quad (3)$$

“” ———

2.2

Theorem 2.2 (——). θ_t *Spring* $\mathcal{M}_t$ “” $\tau \geq 0$ $\theta_{t+\tau} = \theta_t$ $\mathcal{M}_{t+\tau} = \mathcal{M}_t$ $\{E_m\}_{m=1}^M \gamma > 0$ $msq_m(s, \tau) \ell_1$

$$q_m(s, \tau + 1) \leq (1 - \gamma) \cdot q_m(s, \tau) \quad (4)$$

$$\boxed{\lim_{\tau \rightarrow \infty} \Delta_s(t + \tau) = 0} \quad (5)$$

Proof. $\mathbf{1} E_m h_i = \phi_{\theta_t}(s_i) + \text{PE}(p_i)$ $\theta_t \phi_{\theta_t} q_m$ ——
calibration $q_m(s, \tau + 1) \leq (1 - \gamma) q_m(s, \tau)$ “”
 $2M(1 - \gamma)^\tau s$

$$\begin{aligned} \lim_{\tau \rightarrow \infty} p_{\text{clean}, s}(t + \tau) &= \lim_{\tau \rightarrow \infty} \mathbb{E}[\mathbf{1}[E_m(s) \neq y] \mid s] \\ &= P() = \frac{|\mathcal{Y}| - 1}{|\mathcal{Y}|} \end{aligned} \quad (6)$$

$$\begin{aligned} \lim_{\tau \rightarrow \infty} p_{\text{noisy}, s}(t + \tau) &= \lim_{\tau \rightarrow \infty} \mathbb{E}[\mathbf{1}[E_m(s) \neq y] \mid s] \\ &= P() = \frac{|\mathcal{Y}| - 1}{|\mathcal{Y}|} \end{aligned} \quad (7)$$

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$$\lim_{\tau \rightarrow \infty} \Delta_s(t + \tau) = \frac{|\mathcal{Y}| - 1}{|\mathcal{Y}|} - \frac{|\mathcal{Y}| - 1}{|\mathcal{Y}|} = 0 \quad (8)$$

Chernoff-Hoeffding $F_1 \geq 1 - \frac{1}{\eta}$ ——vacuous bound
 $4\gamma \Delta_s(t + \tau) \leq \Delta_s(t) \cdot (1 - \gamma)^\tau$ $T_{1/2} = \log 2 / \log(1/(1 - \gamma)) \approx (\log 2) / \gamma \gamma = 0.110\% T_{1/2} \approx$
6.6 □

□ 1–4 $q_m(s, \tau + 1) \leq (1 - \gamma) q_m(s, \tau)$ Ebbinghaus, 1885 Wixted & Carpenter, 2007 γs ——“”

2.3

Corollary 2.3 (——). $\theta_t s$
 $s \ T_{crit} = O(1/\gamma_s) s \Delta_s$ “”
spaced repetition—— $\Delta_s > 0$ “”

(1) “”——
 “”

(2) $\gamma_s \gamma_s s$ —— $s \ \Delta_s(0) s \ \gamma_s \ \gamma_s$

(3) “” $\tau = 1$ —— $\Delta_s \ \Delta_s$

(4) **bug** γ_s —— $\gamma_s \ \gamma_s = 0$ —— “”

[] “” $q_m \rightarrow 0$ “” —— §6

3 Spring

3.1 Spring SE-1

Spring SE-1(Spring Self-Evolution 1Robbins-Monro)

1: $t < t_{crit} \theta_t K! \ \xi_t \ \|\nabla \mathcal{L}(\theta_t)\|$ —— SCX[?]1.2 $\min_{\tau \leq T} \mathbb{E}[\|\nabla \mathcal{L}(\theta_\tau)\|^2] = O(\log T/\sqrt{T})$

2: $t \geq t_{crit}$ “” Polyak-ŁojasiewiczPLO(1/t) —— $1.1 \mathbb{E}[\mathcal{L}(\theta_t) - \mathcal{L}(\theta^*)] \leq \frac{2L\sigma^2}{\mu^2} \cdot \frac{1}{t}$

Definition 3.1 (——). t_{crit} Spring SE-1

$$t_{crit} = \inf \left\{ t : \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] \leq \epsilon_{basin} \ \lambda_{\min}(\nabla^2 \mathcal{L}(\theta_t)) \geq -\sqrt{\rho \epsilon_{basin}} \right\} \quad (9)$$

$\epsilon_{basin} > 0$ “” “” $\rho > 0$

3.2

Theorem 3.2. Spring $\mathcal{L}L$ - $[\theta^*]$ $\Theta/S_K PLPL\mu > 0 \ t_{crit} < \infty t \geq t_{crit}$

$$\mathbb{E}[\mathcal{L}(\theta_t) - \mathcal{L}(\theta^*)] = O\left(\frac{1}{t}\right) \quad (10)$$

$t < t_{crit} \min_{\tau \leq t} \mathbb{E}[\|\nabla \mathcal{L}(\theta_\tau)\|^2] = O(\log t/\sqrt{t})$

. 1 —— SCX1.2

$$\min_{1 \leq \tau \leq t} \mathbb{E}[\|\nabla \mathcal{L}(\theta_\tau)\|^2] \leq \frac{\mathcal{L}(\theta_0) - \mathcal{L}_{\inf} + \frac{L\alpha^2\sigma^2}{2} H_t}{\alpha \cdot (2\sqrt{t} - 1)} = O\left(\frac{\log t}{\sqrt{t}}\right) \quad (11)$$

“” “/”

2SCX1.3Hessian $\mathbb{E}[T_{\text{escape}}] = O(\log d_{\text{eff}}/(\rho\alpha_t\sigma^2))$

3 —— $\theta_t \theta^* \ \mathcal{NPL}\mathcal{N}\alpha_t = \frac{2}{\mu(t+t_0)} \ 1.1O(1/t)$

41PL $1t_{crit} \theta_t \ t_{crit} = O(K! \cdot \log d_{\text{eff}}/(\rho\alpha\sigma^2)) \ K!$

□

□

[] 1–3SCX1.11.21.3 4PLSpring1 $t_{crit} K!$

3.3

Corollary 3.3 (—). *Spring SE-1* t_{crit}

- (1) $t < t_{crit}$ / — $\Delta_s(t)$ SCX3.1 “” —
- (2) $t \geq t_{crit}$ / “” $O(1/t)$ — $\Delta_s(t)$ PL “” —

- (1) “” $t < t_{crit}$ — “” —
 - (2) “” “” Spring θ_t PL PLO $(1/t)$ $O(\log t/\sqrt{t})$ — “” — Robbins-Monro $\alpha_t \rightarrow 0$
 - (3) $t_{crit} O(K!)$ — K “” $K = 44! = 24$ — $11! = 1$ —
 - (4) “” $\Delta_s(t)$ SCX3.1 $\rightarrow \rightarrow$ “” —
- $\square$ $t_{crit} K!$ — “” KPL PL Spring θ —
Spring SE-1

4

4.1 SCX Theorem 1F1

SCX Theorem 1F1

Theorem 4.1 (SCX Theorem 1 — F1). $M\{E_1, \dots, E_M\} sv_m(s) = \mathbf{1}[E_m(s) \neq y]$ $H_0 v_m \sim \text{Bernoulli}(p_{clean,s})$ $H_1 v_m \sim \text{Bernoulli}(p_{noisy,s})$ $p_{noisy,s} > p_{clean,s}$ $\Delta_s = p_{noisy,s} - p_{clean,s} > 0$
Chernoff-Hoeffding

$$F_1 \geq 1 - \frac{1}{\eta} \sum_{s \in \mathcal{S}} \rho_s \cdot \exp(-2M\Delta_s^2) \quad (12)$$

$$\eta = \sum_s \rho_s \cdot \mathbf{1}[s] \rho_s s$$

$$M\Delta_s F_1 1 - \exp(-2M\Delta_s^2) 1M = 1F_1 1 - \frac{1}{\eta} \sum_s \rho_s \exp(-2\Delta_s^2) —$$

4.2

Corollary 4.2. *SCX Theorem 1*

- (1) $M = 1$ Theorem 1F1 $1 - \frac{1}{\eta} \sum_s \rho_s \exp(-2\Delta_s^2) — \Delta_s \exp(-2\Delta_s^2) \approx 1$ $F_1 1$ “” “”
- (2) $F_1 M$ $F_1 1 - \exp(-2M\bar{\Delta}^2) 1 \bar{\Delta} = \min_s \Delta_s M \geq 3\bar{\Delta} \geq 0.3$ $F_1 \geq 0.95$ — “”
- (3) “” Theorem 1 v_m “” $M_{eff} < M$ — Chernoff M $M_{eff} M$ $M_{eff} \approx 1$ —
- (4) **SATGRE** S Δ_s —
 $M_{eff} = 1$ Theorem 1F1 *vacuous*

- (1) $M = 1$ Theorem 1 — $F_1 \exp(-2M\Delta_s^2)$ formative assessment
- (2) **peer assessment** M $F_1 M$ — rubric M_{eff}

(3) ——— $M = 4$

Theorem 1 $F_1 \Delta_s$ ——— + “” Dunning-Kruger

(4) Theorem 1 F_1

• $\rightarrow$

• $M_{\text{eff}} \rightarrow$

• $\rightarrow$

“+” Theorem 1 F_1

□ Theorem 1 Chernoff-Hoeffding (1) ——— (2) Δ_s ——— (3) Theorem 1

5 ———

5.1

SCXS $i\mathcal{S}_i$

Definition 5.1. $i\mathcal{S}_i \subset \mathbb{R}^{d_i}$

(1) $d_i i d_i$ ———

(2) κ_i “” ——— κ_i

(3) $PL\mu_i\mu_i$ “”

(4) $\gamma_i si$ ——— $s\mathcal{S}_i$

5.2

Theorem 5.2 (———). $N\{1, \dots, N\} \mathcal{S}_i d_i PL\mu_i$ “” $\pi_{uniform} \alpha_t = \alpha/\sqrt{t}$ $i\theta_i^*$

$$\max_{i \in [N]} \mathbb{E}[R_T^{(i)}(\pi_{uniform})] \geq \Omega\left(\max_i d_i \cdot \frac{\log T}{\sqrt{T}} + \frac{1}{\min_i \mu_i} \cdot \frac{1}{T}\right) \quad (13)$$

$\mathcal{S}_i \mu_i PL$

Proof. 1SCX1.2 $\min_{\tau \leq T} \mathbb{E}[\|\nabla \mathcal{L}(\theta_\tau)\|^2] = O(\log T/\sqrt{T})$ d_i ——— $d_i \mathbb{E}[\|\nabla \mathcal{L}(\theta_\tau)\|^2] \geq \Omega(d_i \cdot \log T/\sqrt{T})$

d_i

2PLPLO $(1/\mu_i t) \mu_i$ “” ——— $\alpha_t \alpha_t^{(i)} = 2/(\mu_i(t + t_0^{(i)}))$ ——— $\mu_i t_0^{(i)}$

3 $\propto d_i \log T/\sqrt{T} \propto 1/(\mu_i T) \max_i d_i/\bar{d} \max_i (1/\mu_i)/(1/\mu)$ PL

□

□

□ **SCX1.1–1.2** $1.4K! K_i$ ——— K_i

5.3

Corollary 5.3 (———). *Spring SE-1*

(1) $d_i \mu_i \Omega(\max_i d_i)$ “” $\propto d_{\max}$

(2) “” SCX “” μ_i ——— “” “” ——— $\alpha_t^{(i)} = 2/(\mu_i t)$

(3) $\alpha_t^{(i)} PL\mu_i \mu_i \mu_i +$ “” α_t “” $\alpha_t \mu_i \rightarrow$ “” $\alpha_t \mu_i \rightarrow$

(4) $i d_i$ ——— “” “” ———

- (1) $\mathcal{S}_i \text{ --- } \mu_i \gamma_{s,i}$ “”
 - (2) Khan AcademyALEKS $\Delta_s(t) \text{ --- } \mathcal{S}_i \alpha_t^{(i)}$ Theorem ??
 - (3) “” SCX $\mathcal{S}_i \{ \theta_t^{(i)} \}_{t=0}^T$ “” $\Omega(\max_i d_i)$
- $\square \mathcal{S}_i d_i, \mu_i, \kappa_i, \gamma_{s,i}$ PL $\mu_i \text{ --- } \text{“”}$ Spring SE-1

6

SCX $\square \square \square$

6.1 $\rightarrow$

Table 2:		
SCX		
	$q_m(s, \tau + 1) \leq (1 - \gamma)q_m(s, \tau)$	+ “”
γ	$\gamma s \in \mathcal{S}$	vs vs
	$q_m \rightarrow 0 \implies \Delta_s \rightarrow 0$	
/		Δ_s

“” “” savings effect ---
 SCX $r_s(\tau) \text{ --- } \Delta_s \rightarrow 0 \text{ } r_s > 0$ “” “”

6.2 $\rightarrow$ PL

PLPolyak-Łojasiewicz $\frac{1}{2} \|\nabla \mathcal{L}(\theta)\|^2 \geq \mu(\mathcal{L}(\theta) - \mathcal{L}(\theta^*))$
 Liu et al., 2022 “” $\mathcal{L}_{\text{human}}(\theta)$ “ ” Spring SE-1
 $t_{\text{crit}} = O(K! \cdot \log d_{\text{eff}}/(\rho \alpha \sigma^2))$ $K = 5 \text{ } 120 \times \text{ --- }$

6.3 $\rightarrow$

- (1) “” $M_{\text{eff}} M_{\text{nominal}}$
- (2) “” “” “” --- Theorem 1Chernoff --- F_1 “ M_{eff} ” “”
- (3) --- Δ_s testing effect Δ_s Theorem 1
- (4) Theorem 1 “” y
 “” “” --- “” “”

“” “reflective equilibrium” ---
 $\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{m=1}^M v_m(s)$ “” p_s

6.4 $\rightarrow$

$\mathcal{S}_i d_i \mu_i$ PL κ_i “” ---
Theorem 3 --- $\mu_i \Delta_s$ /

Table 3:			
SCX			
	Spring SE-1	$\square$	
$>$	Theorem 1	$\square$	
		$\square$	$\mathcal{S}_i$

6.5

SCX possibility proof—— —— falsifiable hypothesis

7

7.1 SCX

SCX

- (1) Spring SE-1——
- (2) $F_1\exp(-2M\Delta_s^2)$ Theorem 1—— Chernoff
- (3) $\rightarrow$ ——
- (4) ——

SCX

- (1)
- (2) $KPL\mu$
- (3) “”Theorem 1
- (4) effect size——

7.2

VygotskyZPDSCXZPD $\|\nabla\mathcal{L}(\theta)\|$ —— scaffolding PL——Vygotsky“”
 BloomMastery LearningBloom 80%SCX $\Delta_s(t) \geq \tau_{\text{mastery}}\mathcal{S} \Delta_s\tau_{\text{mastery}} \Delta_s$ ——
 SwellerCLTIntrinsic extraneousgermaneSCX “”—— Spring SE-1

7.3 Open Problem

- (1) PLDuolingo Khan AcademyPLO(1/t) PL
- (2) γ_s SCX s γ_s
- (3) M_{eff}
inter-rater reliability
- (4) Spring SE-1Spring“”——
SCX“”
- (5) SCX—— SCXSpring
 M
- (6) Theorem 3SCXTheorem 3 “” “”—— ——

7.4 ———

———— SCX
 Ebbinghaus BloomSCX——
 ——— $\gamma > 0$ PL $M_{\text{eff}} \geq 3\mathcal{S}_i$ ——— ———
 SCXSpring SE-1 “”——Spring SE-1 ———

——Spring SE-1Theorem 1F1—— SCX[?, ?, ?, ?] SCX SCX[?]——

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